D 30179 (Pages : 4) Name.....

Reg. No.....

FIFTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION NOVEMBER 2022

Mathematics

MAT 5B 07—BASIC MATHEMATICAL ANALYSIS

(2017—2018 Admissions)

Time : Three Hours Maximum : 120 Marks

Part A

Answer all the **twelve** *questions.*
Each question carries 1 *mark.*

1. Fill in the blanks : Infimum of the set $S = \{1/3^m - 1/4^n \; ; m, n \in N\}$ is .....
2. The Set of all real numbers which satisfy the inequality $|x-1| < |x-3|$ is .....
3. The Supremum property of R states that .....
4. State Bernoulli's Inequality.
5. State Cantor's theorem .
6. Give an example of a bounded real sequence which is not monotone.
7. State the Characterization theorem of open sets.
8. Fill in the blanks : $\lim\left(\frac{\sin n}{n}\right) =$ ————.
9. If $b > 0$, then $\lim\left(\frac{1}{1+nb}\right) =$ ————.
10. Define contractive sequence.
11. Fill in the blanks : Arg (-2π) = ————.
12. The value of $(1+i)^{24}$ is ————.

(12 × 1 = 12 marks)

Turn over

Part B

Answer any **ten** *questions.*
Each question carries 4 *marks.*

13. Define Supremum and Infimum of set . Find them for the Set $S = \left\{1 - \frac{(-1)^n}{n}; n \in N\right\}$.

14. If $x > -1$, then show that $(1+x)^n \geq 1 + nx, \forall n \in N$.

15. If $y > 0$, then show that there exist some $n_y \in N$, such that $n_y - 1 \leq y < n_y$.

16. Show that there doesnt exist a real number r such that $r^2 = 5$.

17. If x and y are real numbers satisfying $x < y$, then prove that there exist an irrational number z such that $x < z < y$.

18. State the completeness property of R. Prove or disprove that the sub field Q of rational numbers has this property .

19. Define the limit of a sequence. Using the definition of limit, prove that $\lim (1/n) = 0$.

20. State and prove Squeeze theorem for sequence .

21. If $X = (x_n)$ is a converging sequence of non-negative real numbers then prove that $\lim (x_n) \geq 0$.

22. Show that every Cauchy sequence of real numbers is bounded.

23. Let $X = (x_n)$ and $Y = (y_n)$ be real sequences that converges to x and y respectively. Prove that $X + Y$ converges to $x + y$.

24. Define contractive sequence. Test whether the sequence $X = (x_n)$ defined by $x_1 = 2$, $x_{n+1} = 2 + 1/x_n$, $n \in N$ is a contractive sequence or not.

25. Find a sequence (x_n) of real numbers such that $\lim \left| x_{n+1} - x_n \right| = 0$, but not a Cauchy sequence.

26. Find the Arg Z, if $Z = \left(\sqrt{3} + i\right)^6$.

(10 × 4 = 40 marks)

Part C

Answer any **six** *questions.*

Each question carries 7 *marks.*

27. Show that the unit interval [0, 1] is uncountable.

28. Determine the set $A = \left\{ x \in R : \frac{2x+1}{x+2} < 1 \right\}$.

29. State and prove the characterization theorem of intervals.

30. State and prove the "Ratio test " for sequence.

31. If $0 < a < b$, then test the convergence of the sequence (x'_n), where $x_n = \frac{a^{n+1} + b^{n+1}}{a^n + b^n}$.

32. Test the convergence of (x_n) defined by $x_n = 1 + 1/2! + 1/3! + \ldots\ldots + 1/n!$

33. Prove that a subset of R is closed if and only if it contains its cluster points.

34. Find all values of $(-216\, i)^{\frac{1}{3}}$.

35. Factorise $x^6 - 1$ into linear factors.

(6 × 7 = 42 marks)

Part D

Answer any **two** *questions.*

Each question carries 13 *marks.*

36. (a) State and prove the "Density theorem " of rational numbers.

(b) Discuss the convergence of the following sequences whose n^{th} terms are defined by :

(i) $x_n = \left(1 + \frac{1}{n^2}\right)^{2n^2}$; and (ii) $y_n = \frac{\log n}{n}$.

Turn over

37. (a) State and prove the Cauchy convergence criterion for sequence.

    (b) Discuss the convergence of X = (x_n) , defined by $x_n = \ln n$.

38. (a) Define closed sets in R show that the Intersection of an arbitrary collection of closed sets in R is closed.

    (b) Show by an example that the union of infinitely many closed sets in R need not be closed.

(2 × 13 = 26 marks)